Graphs

CT0540 - Social network analysis

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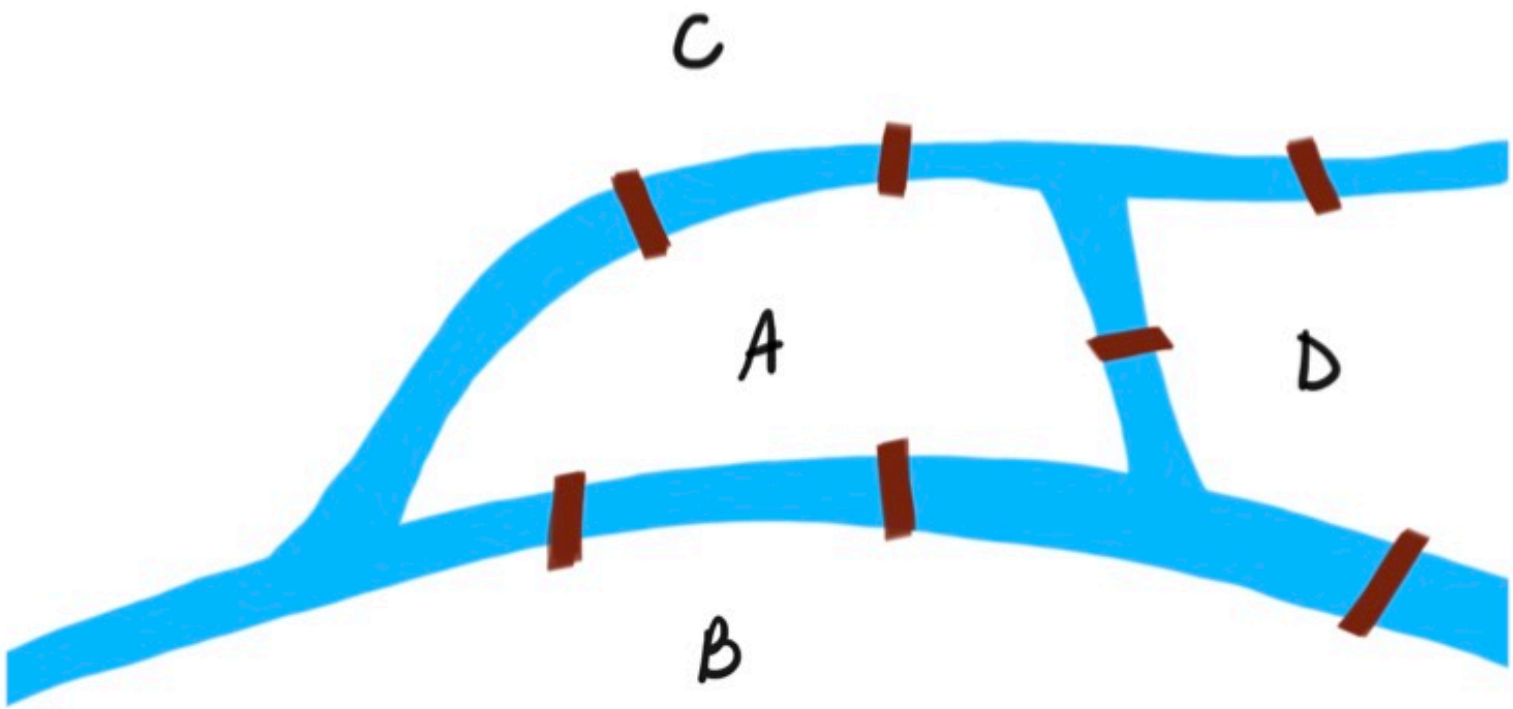
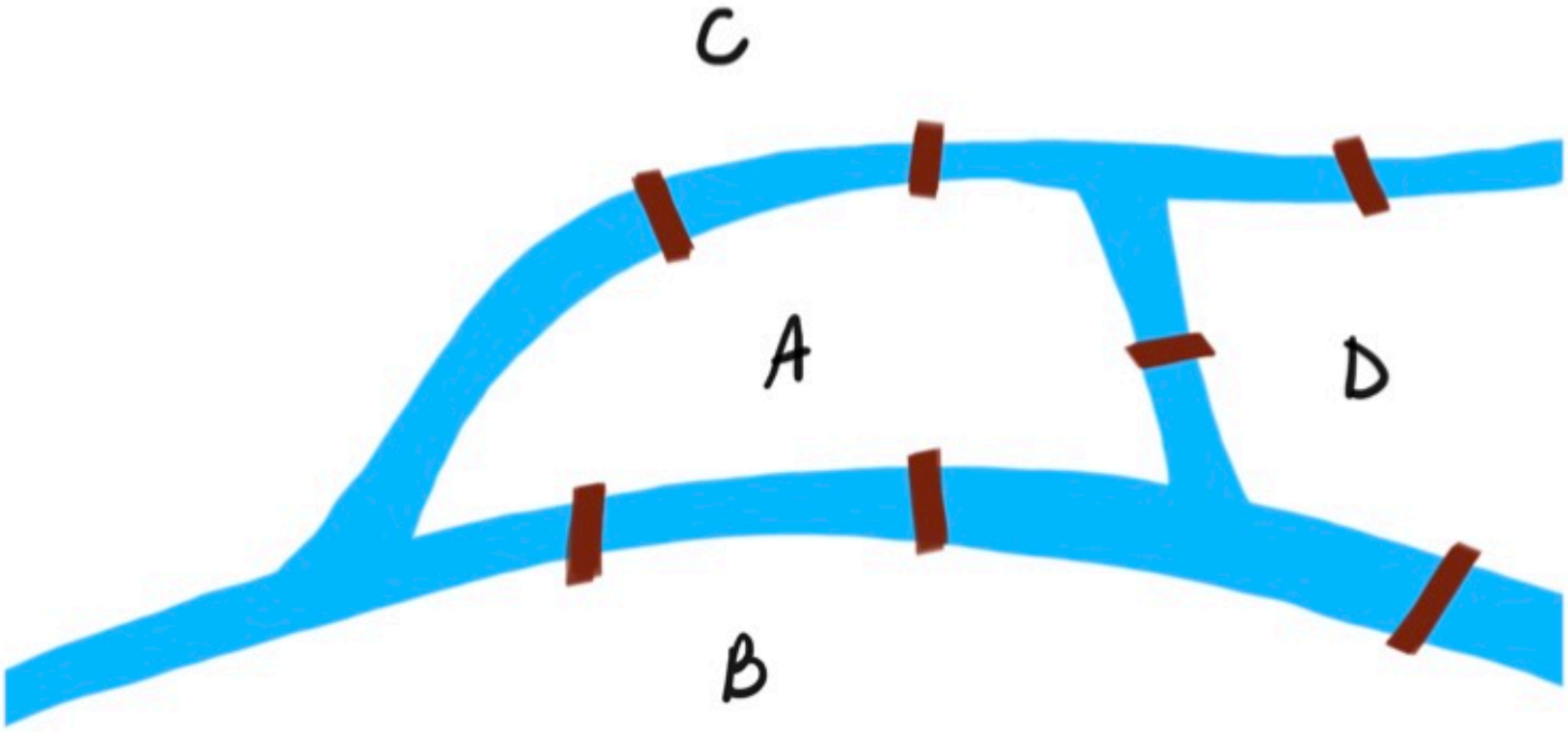
Introduction

- 1735 in Koenigsberg, the capital of Eastern Prussia, a merchant city of its time
- There were seven bridges across the river Pregel that surrounded the town



The Bridges of Koenigsberg

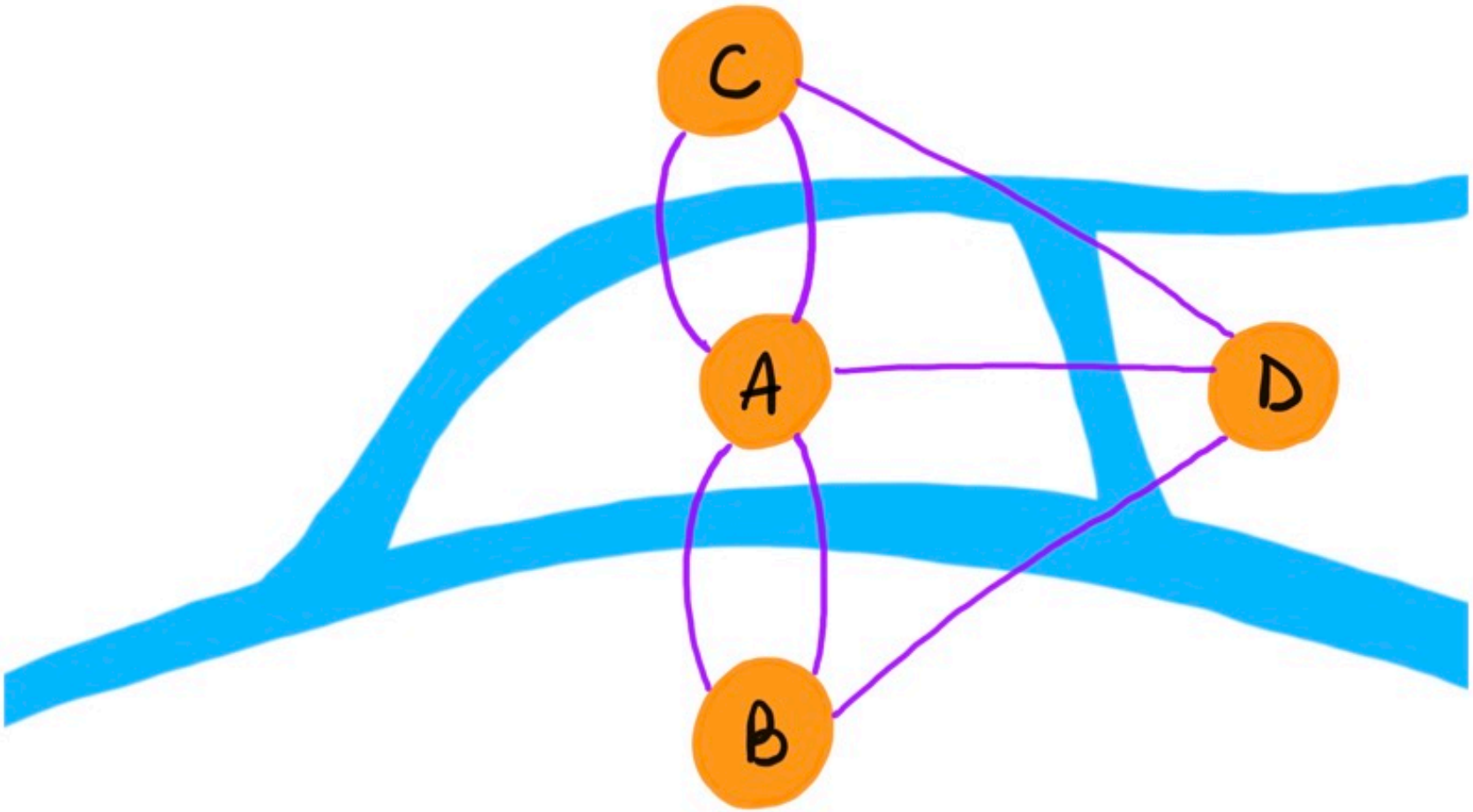
Can one walk across all seven bridges and never cross the same one twice?



The Bridges of Koenigsberg

- The problem was solved by Leonard Euler, a mathematician
- Euler constructed a graph that has
  - four nodes (A, B, C, D), each corresponding to a patch of land
  - seven links, each corresponding to a bridge

The Bridges of Koenigsberg



The Bridges of Koenigsberg

- Euler offered a mathematical proof that such path does not exist
- There is no continuous path that would cross the seven bridges while never crossing the same bridge twice

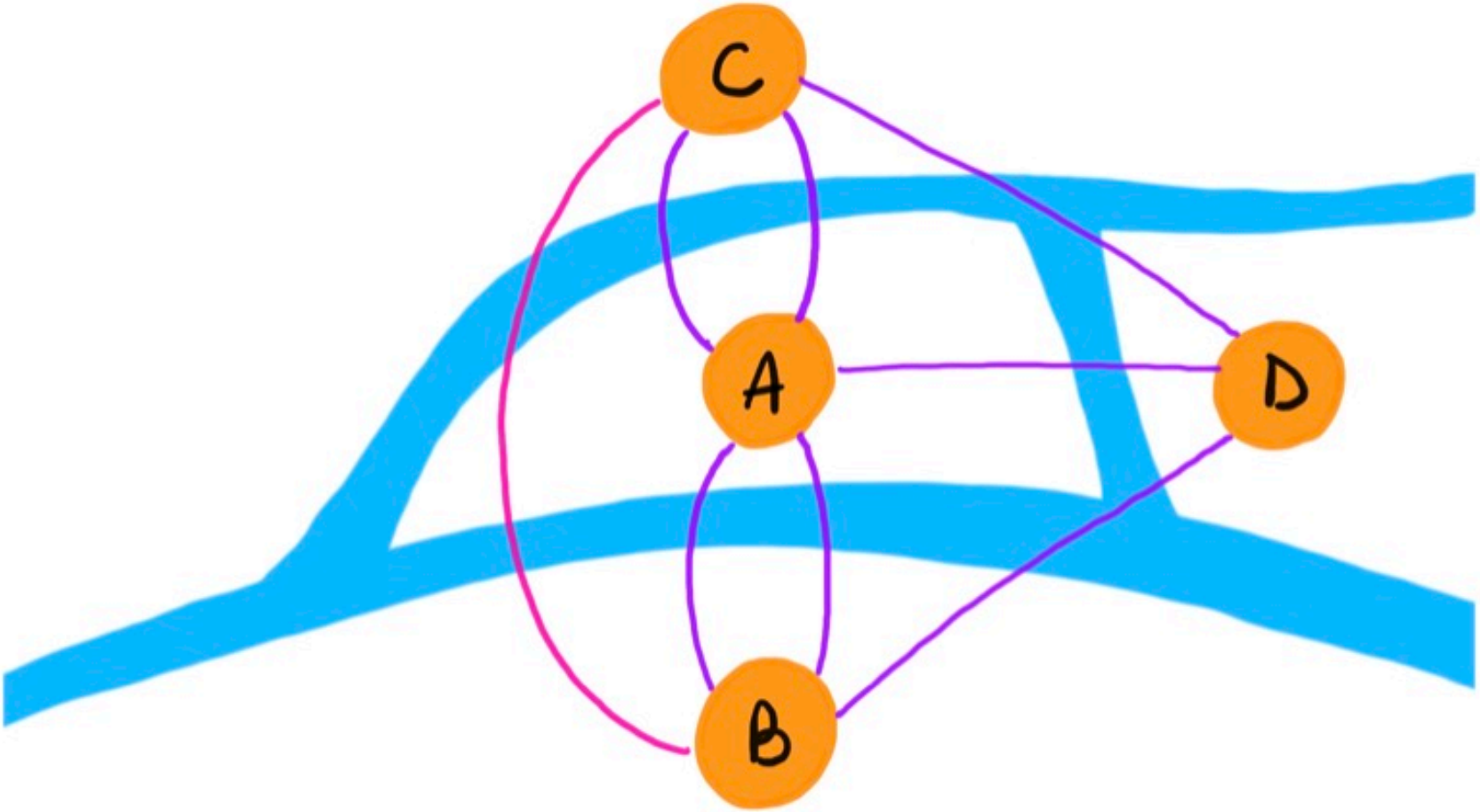
The Bridges of Koenigsberg

- A walking path that goes through all bridges can have only one starting and one end point
- Such a path cannot exist on a graph that has more than two nodes with an odd number of links
- The Koenigsberg graph had four nodes with an odd number of links
  - No path could satisfy the problem!

The Bridges of Koenigsberg

- The people of Koenigsberg in 1875 built a new bridge between B and C
- Now only one node was left with an odd number of links
- And we thus should be able to find the desired path!

The Bridges of Koenigsberg



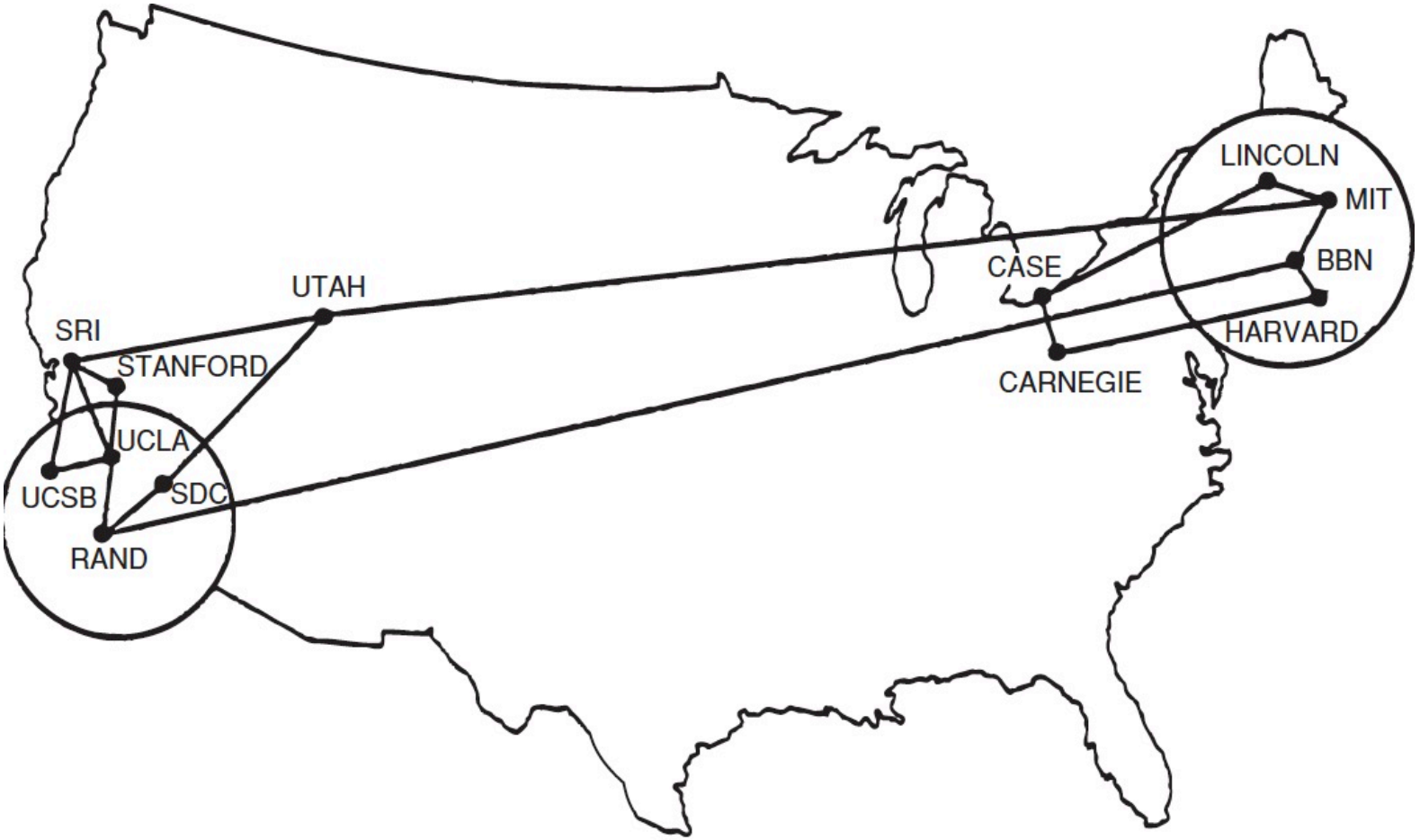
The Bridges of Koenigsberg

- Euler's proof was the first time someone solved a mathematical problem using a graph
- Two key messages:
  - Some **problems become simpler** and more tractable if they are represented as a graph
  - Graph have **properties** that limit or enhance their behavior

Video: The Koenigsberg's problem and Euler's solution

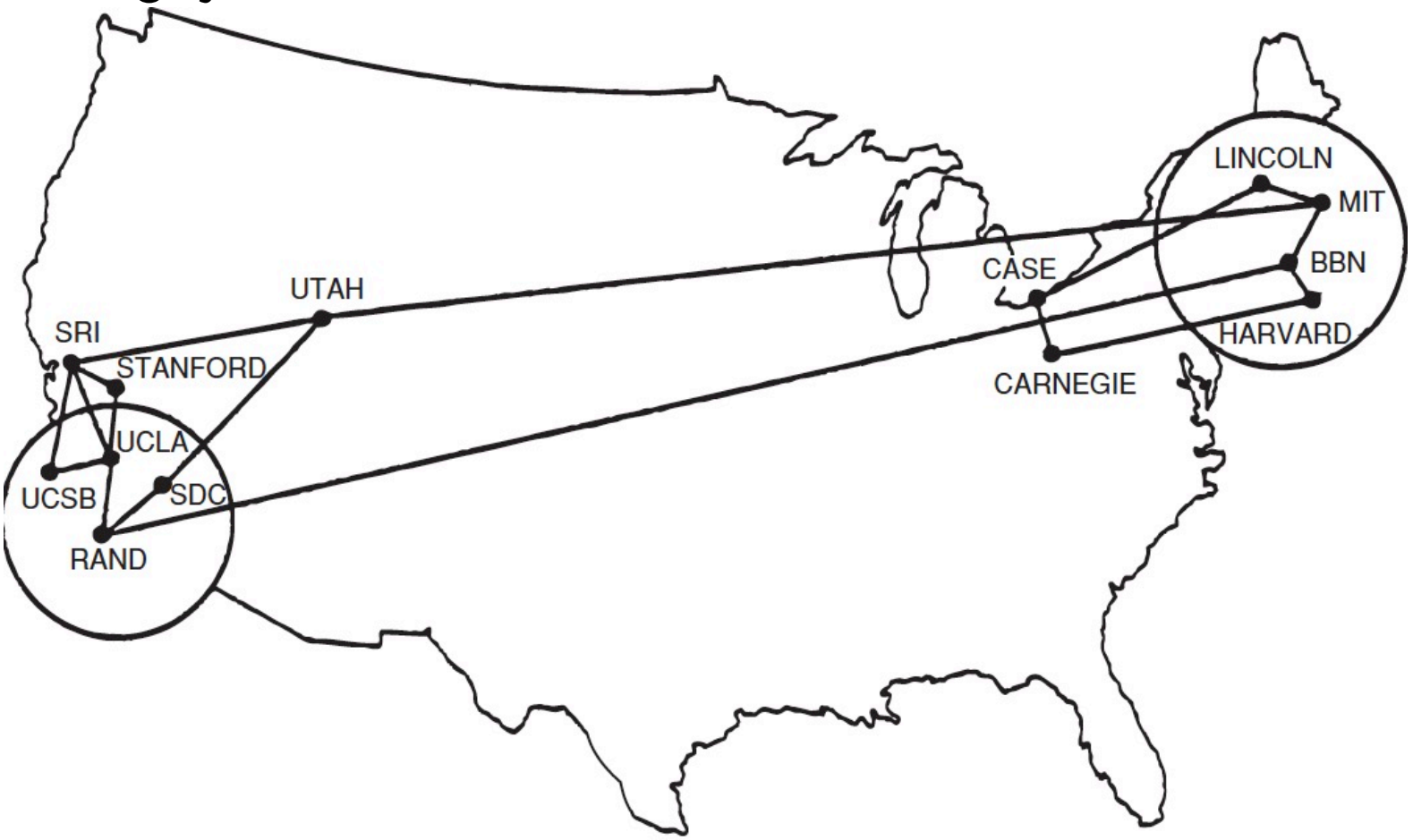
Graphs as Models of Networks

- Graphs are useful because they serve as mathematical models of network structures
- The Figure shows the network structure of the Internet (ARPANET) in 1970



Graphs as Models of Networks

- Nodes are computing hosts
- An edge joins two nodes if there is a direct communication link



Graphs as Models of Networks

- Nodes are computing hosts
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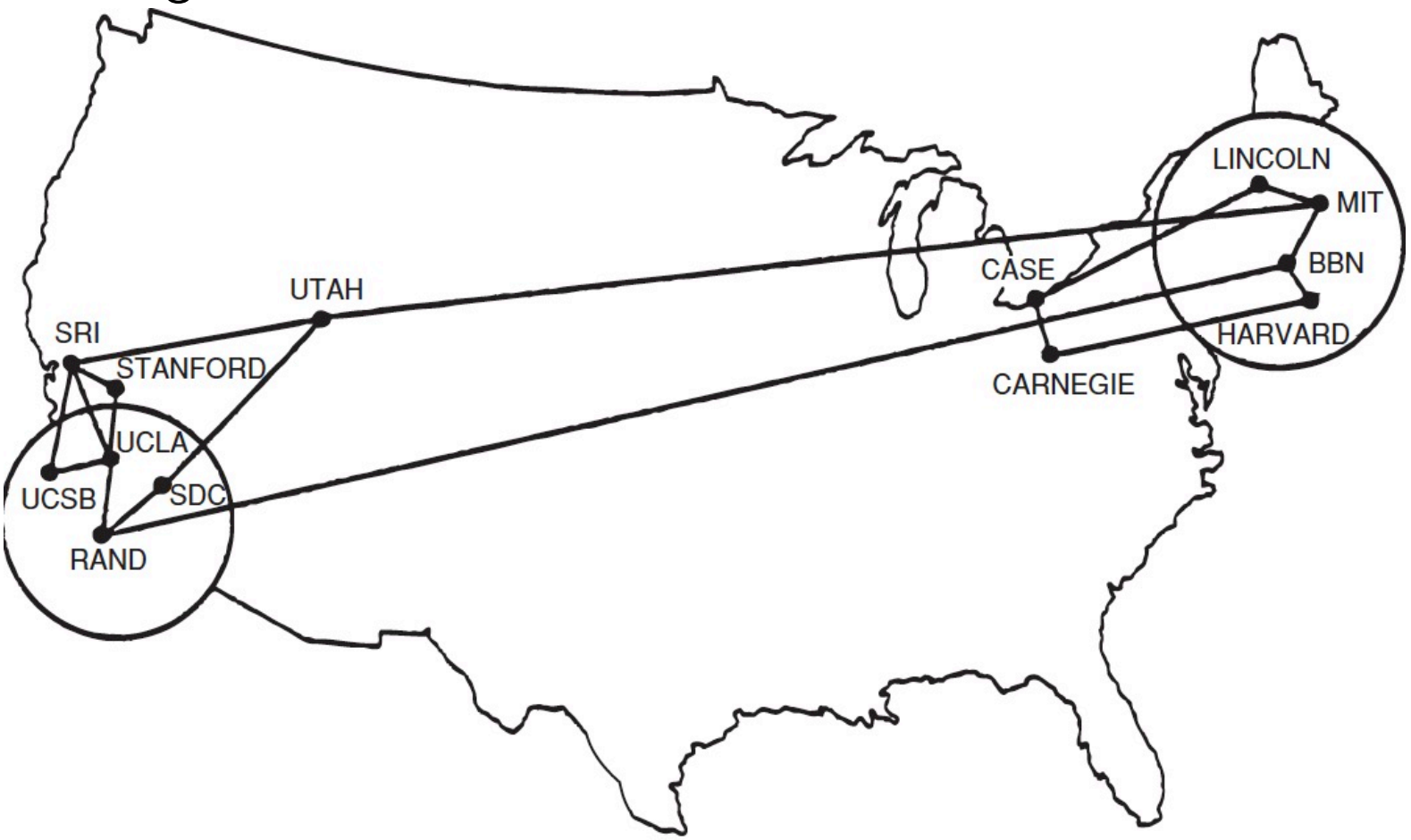
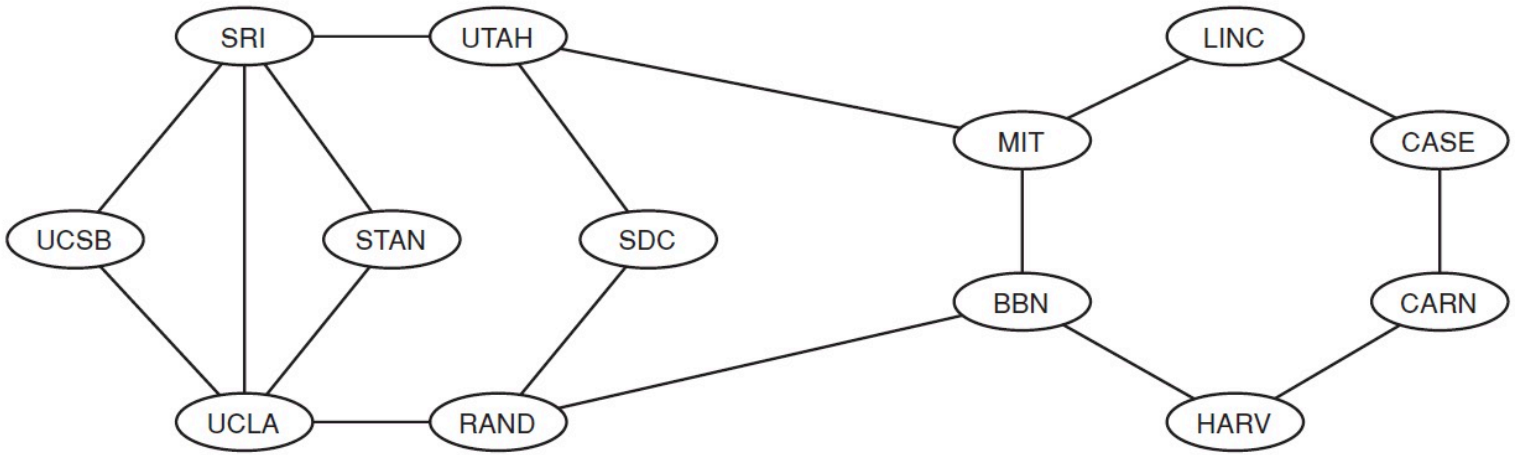
- Note that the actual placement or layout of the nodes is meaningless!



Graphs as Models of Networks

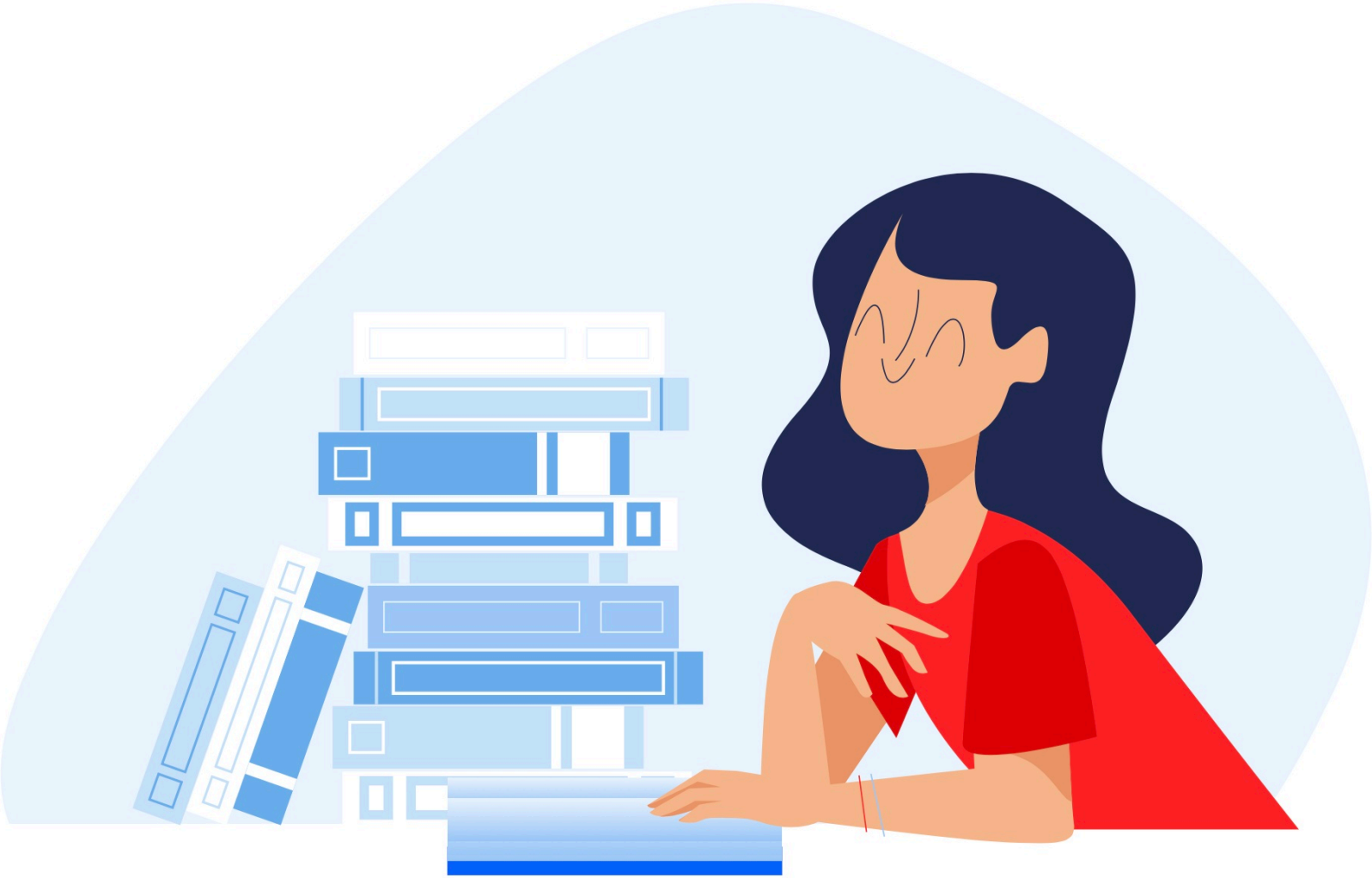
- Note that the actual placement or layout of the nodes is meaningless!
- The figure below shows an alternate drawing of the same graph



Graphs as Models of Networks

- ARPANET is an example of **communication networks**, in which nodes are devices and edges represent direct links along which messages can be transmitted
- In **information networks**, nodes are information resources (web pages, documents) and edges are logical connections (hyperlinks, citations)
- In **social networks**, nodes are people or groups of people, and edges represent some kind of social interaction

Properties and fundamental concepts



Degree and Average Degree

- A key property of each node is its degree, representing the number of links it has to other nodes
- Node *i* has degree k_i
- In an undirected network, the total number of links L is given by:

$$L = \frac{1}{2} \sum_{i=1}^{N} k_i$$

Since in the sum each link is counted twice, we need the 1/2 factor

Degree and Average Degree

- Another important property is the average degree
- For an **undirected network** it is

$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^{N} k_i = \frac{2L}{N}$$

Degree and Average Degree

- For directed networks, we distinguish between
  - **incoming degree**, the number of links pointing to node *i*
  - **outgoing degree**, the number of links pointing from node *i* to other nodes
- A node's total degree is thus given by

$$k_i = k_i^{in} + k_i^{out}$$

Degree and Average Degree

The total number of links in a directed network is

$$L = \sum_{i=1}^{N} {k_i}^{in} = \sum_{i=1}^{N} {k_i}^{out}$$

The average degree of a directed network is

$$\langle k^{in} \rangle = \frac{1}{N} \sum_{i=1}^{N} k_i^{in} = \frac{L}{N} = \frac{1}{N} \sum_{i=1}^{N} k_i^{out} = \langle k^{out} \rangle$$

Degree Distribution

The **degree distribution** provides the probability that a randomly selected node has degree k and is given by

$$p_k = \frac{N_k}{N},$$

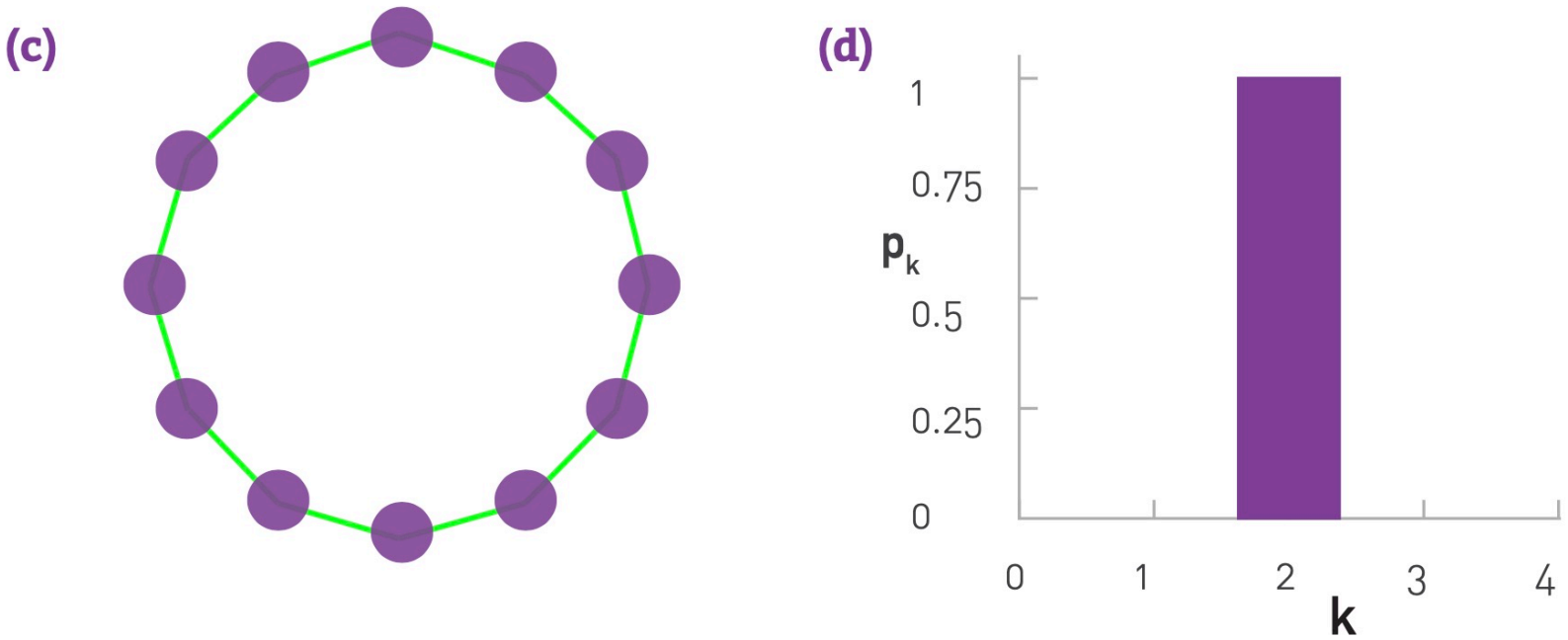
where N_k is the number of degree-k nodes.

The number of degree-k nodes can be obtained from the degree distribution as

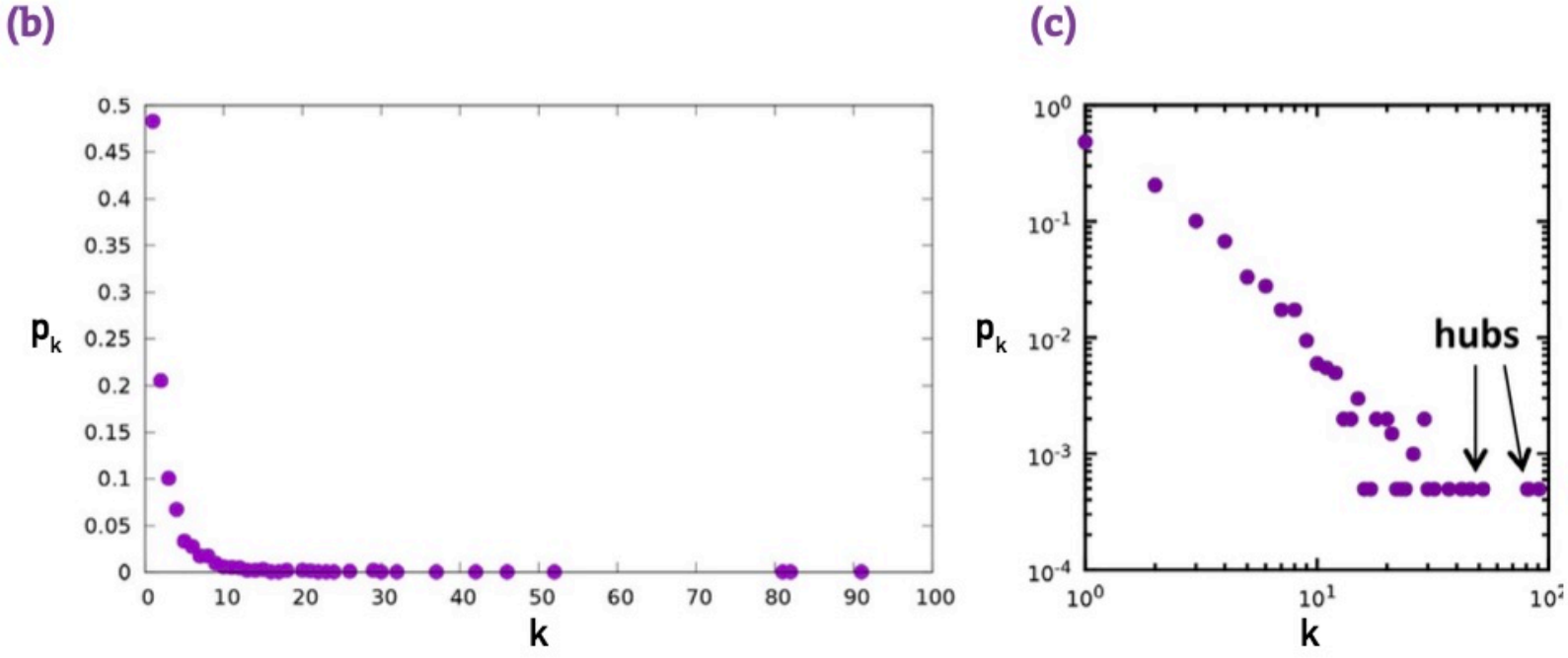
$$N_k = N p_k$$

Degree Distribution

Degree Distribution



Degree Distribution of a Real Network



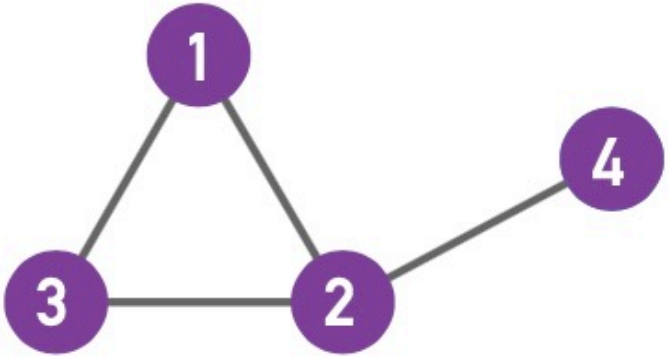
Adjacency Matrix

- It's useful to describe a network through a list of its links
- The **adjacency matrix** of a directed network of N nodes has N rows and N columns
  - $A_{ij} = 1$ if there is a link from node j to node i
  - $A_{ij} = 0$ if nodes i and j are not connected

$$A_{i,j} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N} \end{pmatrix}$$

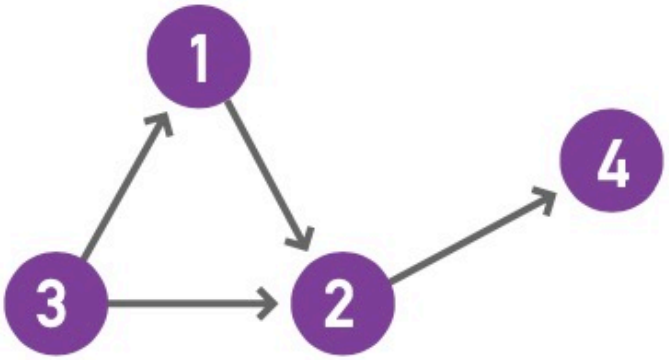
Adjacency Matrix

(b) Undirected network



$$A_{ij} = \begin{matrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

(c) Directed network



$$A_{ij} = \begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

Adjacency Matrix

- The degree of a node can be directly obtained from the elements of the adjacency matrix
- For **undirected networks**:

$$k_i = \sum_{j=1}^{N} A_{ji} = \sum_{i=1}^{N} A_{ji}$$

- For **directed networks**:
  - Incoming degree: $k_i^{in} = \sum_{j=1}^{N} A_{ij}$
  - Outgoing degree: $k_i^{out} = \sum_{j=1}^{N} A_{ji}$

Adjacency Matrix

- G is undirected $\iff$ A is symmetric
- G has a self-loop $\iff$ A has a non-zero element in the diagonal
- G is complete $\iff$ $A_{ij} \neq 0 \quad \forall i \neq j$

Real Networks are Sparse

- Theoretically:

$$|E|_{max} = \binom{N}{2} = \frac{N(N-1)}{2}$$

- In real life, most real networks are **sparse**, i.e.,

$$|E| \ll |E|_{max}$$

Real Networks are Sparse

- How sparse is Zachary's Karate Club's network?
  - $|V| = 34, |E| = 78, |E|_{max} = 561$
- Sparseness has important consequences on how we **explore** and **store** real networks
- Example: when we store a large network in our computer, it is better to store only the list of links, rather than the full adjacency matrix

Weighted Networks

- Sometimes we need to study weighted networks
- Each link has a weight
- For weighted networks the elements of the adjacency matrix carry the weight of the link as

$$A_{ij} = w_{ij}$$

Weighted Networks

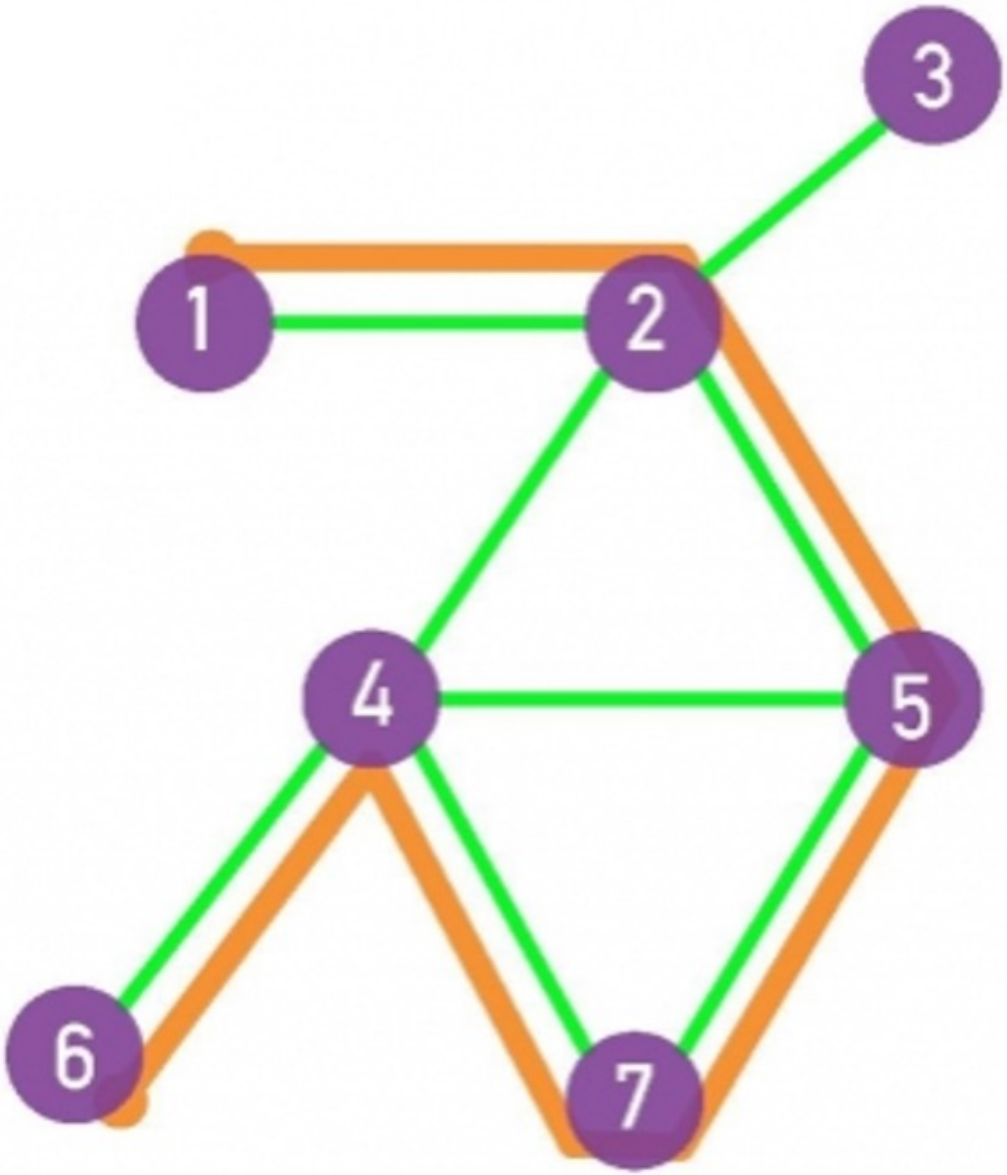
- Example:
  - In a mobile call network, the weight can represent the total number of minutes two individuals talk with each other on the phone
  - In a friendship network, the weight can represent the total number of years two friends have known each other

Paths

- A path is a **sequence of edges**
- The destination of each edge is the origin of the next edge
- The **length of the path** is the number of edges on it

Paths

Example: A path of length 5 (orange)

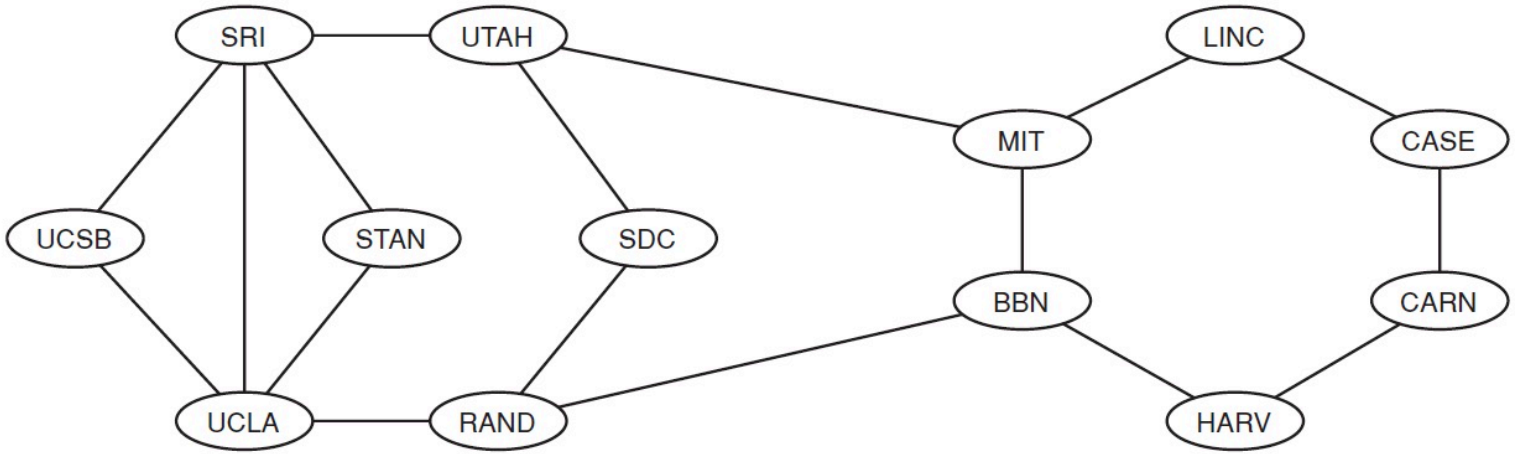


Paths

- An important kind of path is a **cycle**, which is a path with at least 3 edges, in which the first and last nodes are the same, but all other nodes are distinct
- Informally, a cycle is a **ring** structure

Paths

- In the ARPANET graph, there are many cycles:



Paths

- In ARPANET, cycles are there by design: if any edge were to fails, there would still be a way to get from any node to any other node in the network
- In transportation and communication networks, cycles are often present to allow for **redundancy**
  - They provide alternative routes!

Paths

- What about cycles in social networks?

Paths

- What about cycles in social networks?
- In a social network of friendships, there are often cycles, e.g.,
  - Your wife's cousin's close friend from high school works with your brother
  - You - Wife - Cousin - Friend - Brother - You

Connectivity

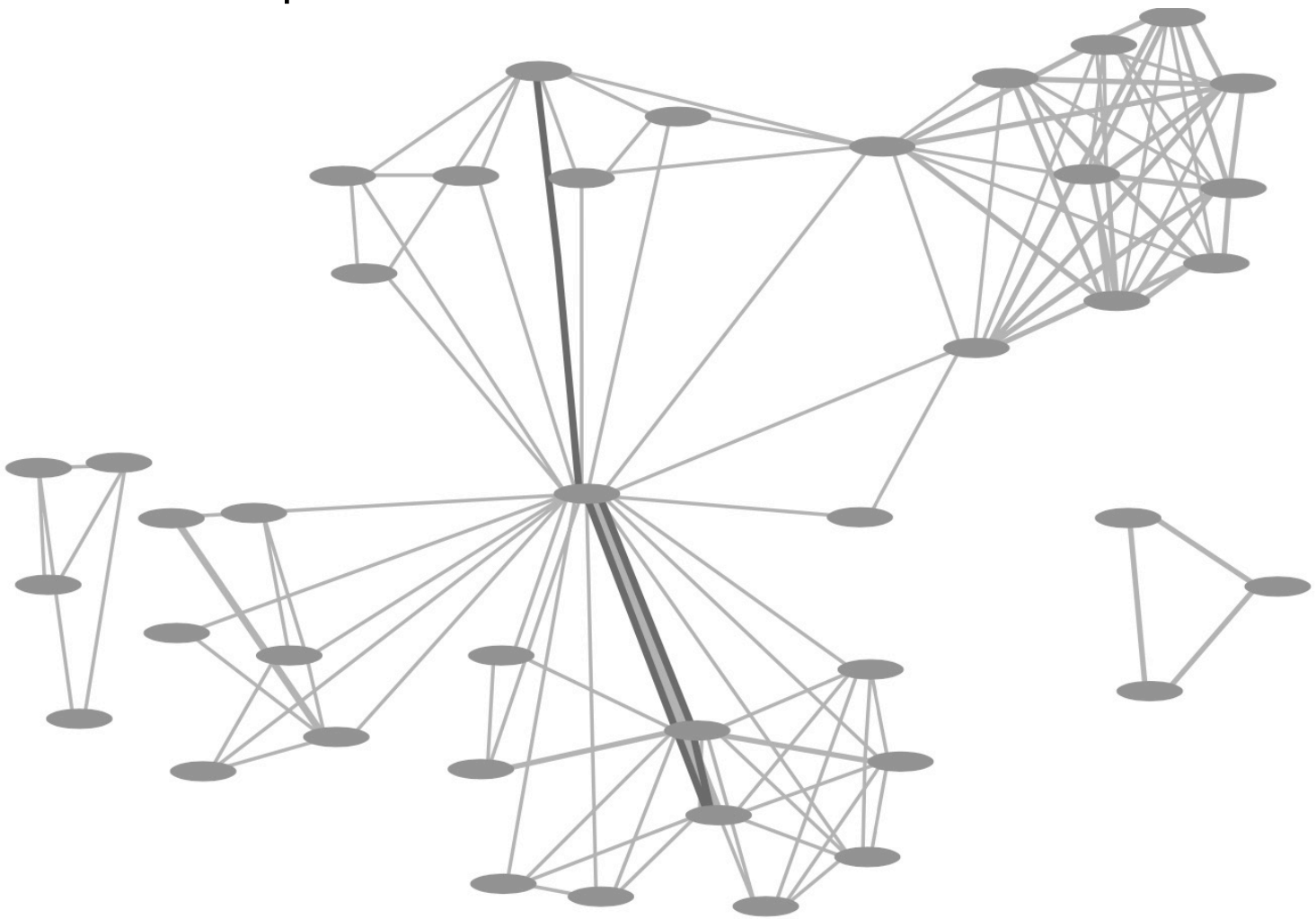
- Can every node in the graph reach every other node by a path?
- If yes, we say that the graph is **connected**
- In other words, for every pair of nodes, there is a path between them
- As we would expect, the ARPANET graph is connected (the goal is to move traffic between nodes)

Connectivity

- In other settings, graphs are not necessarily connected
- For example, in social networks, there may exist two people who are not connected to one another
- Let's see an example..

Connectivity

- Collaboration graph at a biological research center
- Nodes are researchers, edges exist if two researchers co-authored a publication



Connectivity

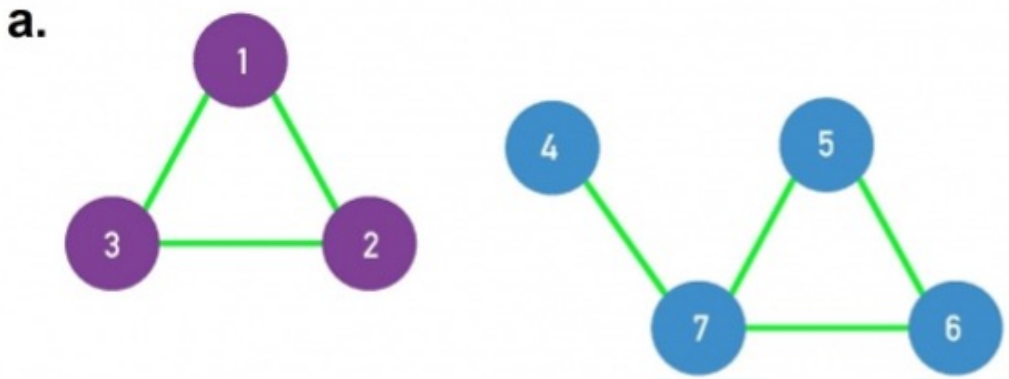
- If a path exists between two nodes i and j
  - These nodes are part of the same **connected component**
- A graph that has only one connected component is called a **connected graph**

Connectivity

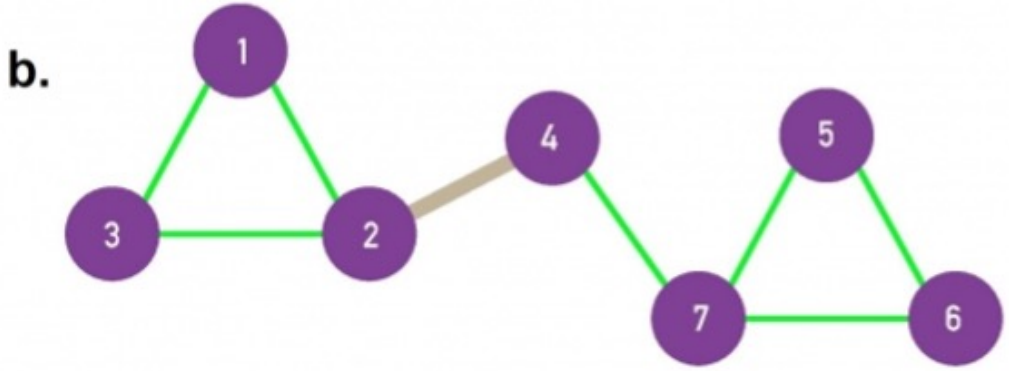
- A **connected component** of a graph is a subset of nodes such that:
  1. Every node in the subset has a path to every other **AND**
  2. The subset is not part of a larger set with the property that every node can reach any other
- So we have two conditions:
  1. The component is internally connected
  2. The component is not connected to a larger piece of the graph

Connectivity

- A connected graph has only one connected component
- In a disconnected graph the adjacency matrix can be arranged in block diagonal form



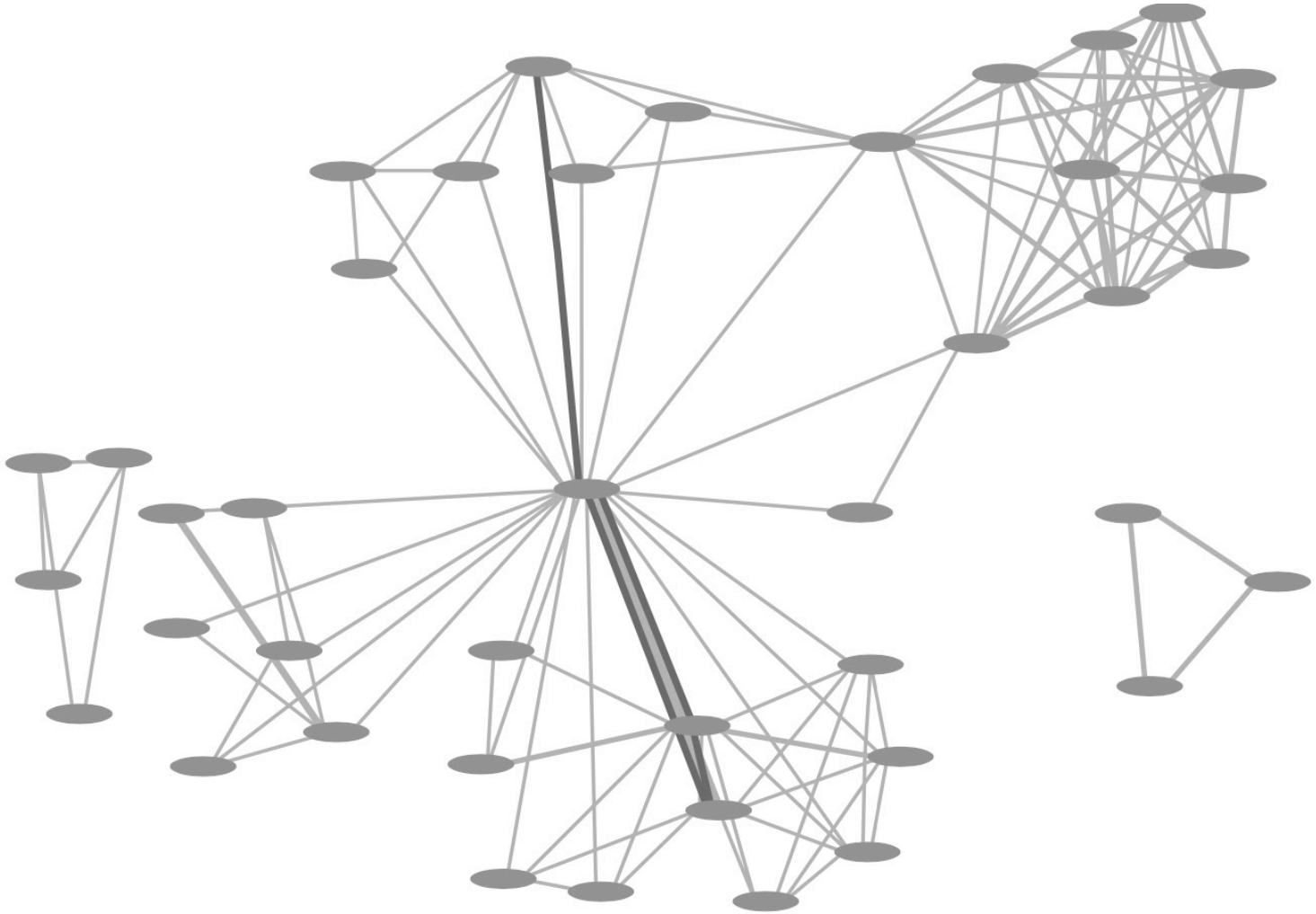
$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$



$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

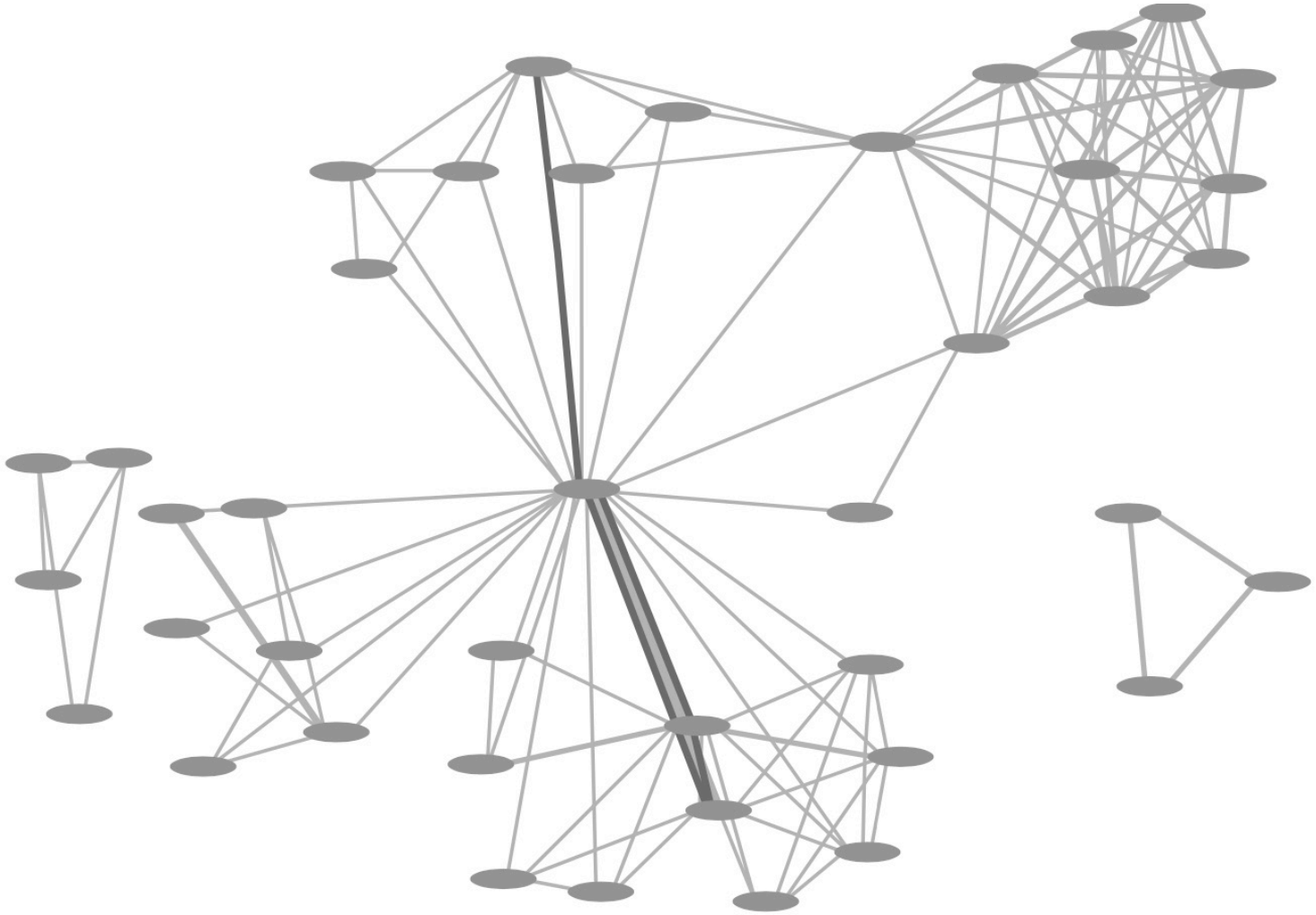
Connectivity

- Let's go back to the example of the collaboration graph..
- .. what do you notice?



Connectivity

- We have a prominent node at the center and groups linked to this node but not to each other
- The largest connected component would break into 3 distinct components if that node were removed



Giant Components

- Consider the social network of the entire world, with a link between two people if they are friends
- Is this global friendship network connected?

Giant Components

- Consider the social network of the entire world, with a link between two people if they are friends
- Is this global friendship network connected?
- Probably not

EUROPE
ICELAND
NORTH SEA
BRITISH ISLES
London
GERMANY
POLAND
Berlin
FRANCE
Paris
SPAIN
Madrid
Biscay
Bay of
ITALY
Rome
Adriatic Sea
YUGOSLAVIA
ALGERIA
LIBYA
EGYPT
Cairo
Tripoli
Morocco
Fez
Gibraltar
New York
Black Sea
Caspian Sea
Mediterranean Sea
Sicily
Corsica
Sardinia
Malta
Crete
Cyprus
Rhodes
Ankara
Tehran
Baghdad
Damascus
Jerusalem
Beirut
Athens
Sofia
Bucharest
Budapest
Vienna
Prague
Warsaw
Moscow
Leningrad
Stockholm
Oslo
Copenhagen
Dublin
Lisbon
Madrid
Paris
Rome
Cairo
Tehran
Moscow
Berlin
London
New York

Giant Components

- And yet.. Let's consider an example
- I have friends in other countries
  - We are in the same component
- The parents of these friends, and their friends, and their descendants, are in the same component as well!
- But we never met and probably have very different backgrounds

Giant Components

- Even though the global friendship network is not connected, my component is very large, reaching most parts of the world
- This is often observed in large complex networks
- A **giant component** is a connected component that contains a significant fraction of all nodes

Giant Components

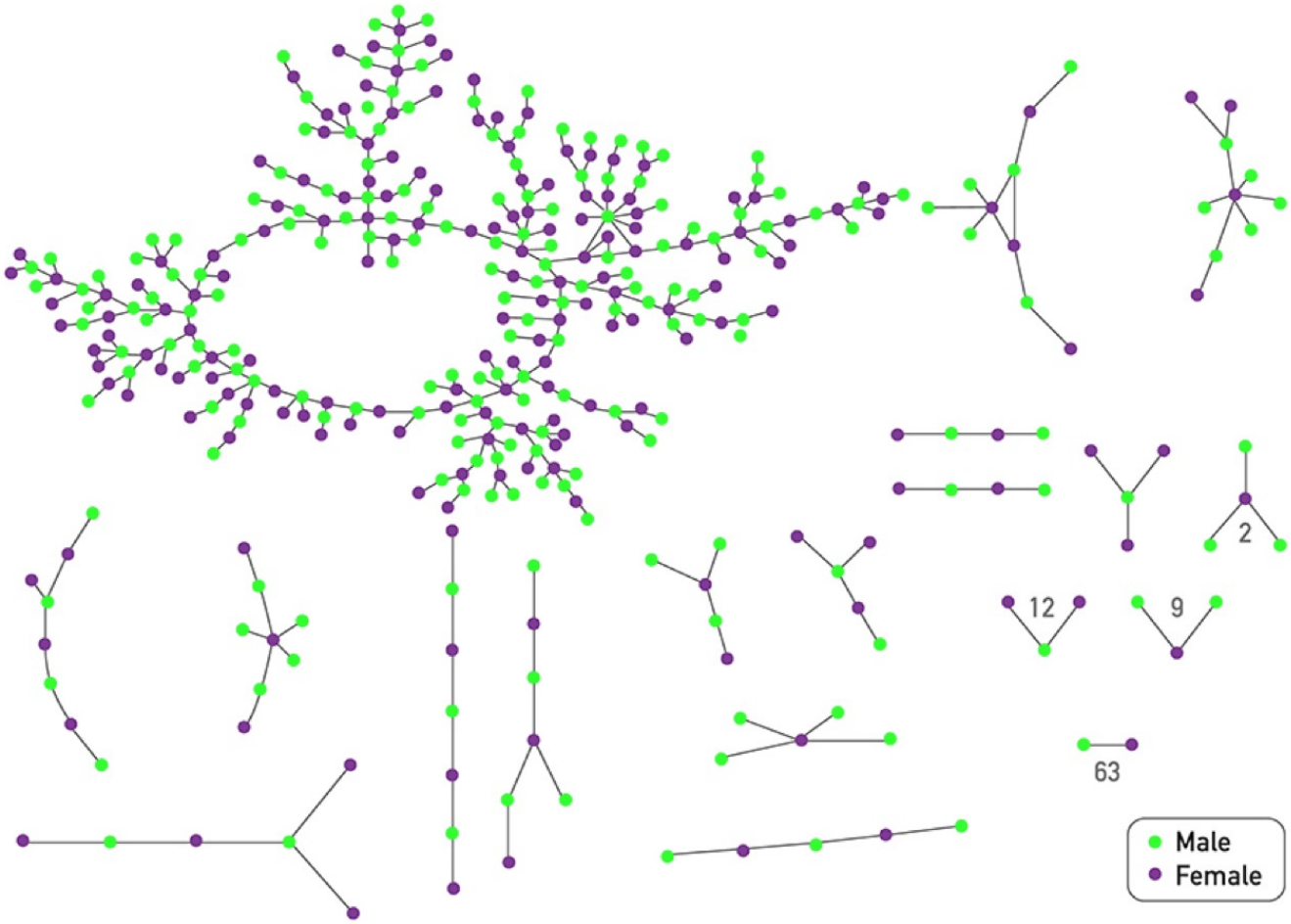
- When a giant component exists in a network, it is usually the only one. Why?

Giant Components

- When a giant component exists in a network, it is usually the only one. Why?
- Let's imagine to have two giants components in the global friendship networks
- It would be sufficient to have a single edge to merge the two components and usually such an edge would form

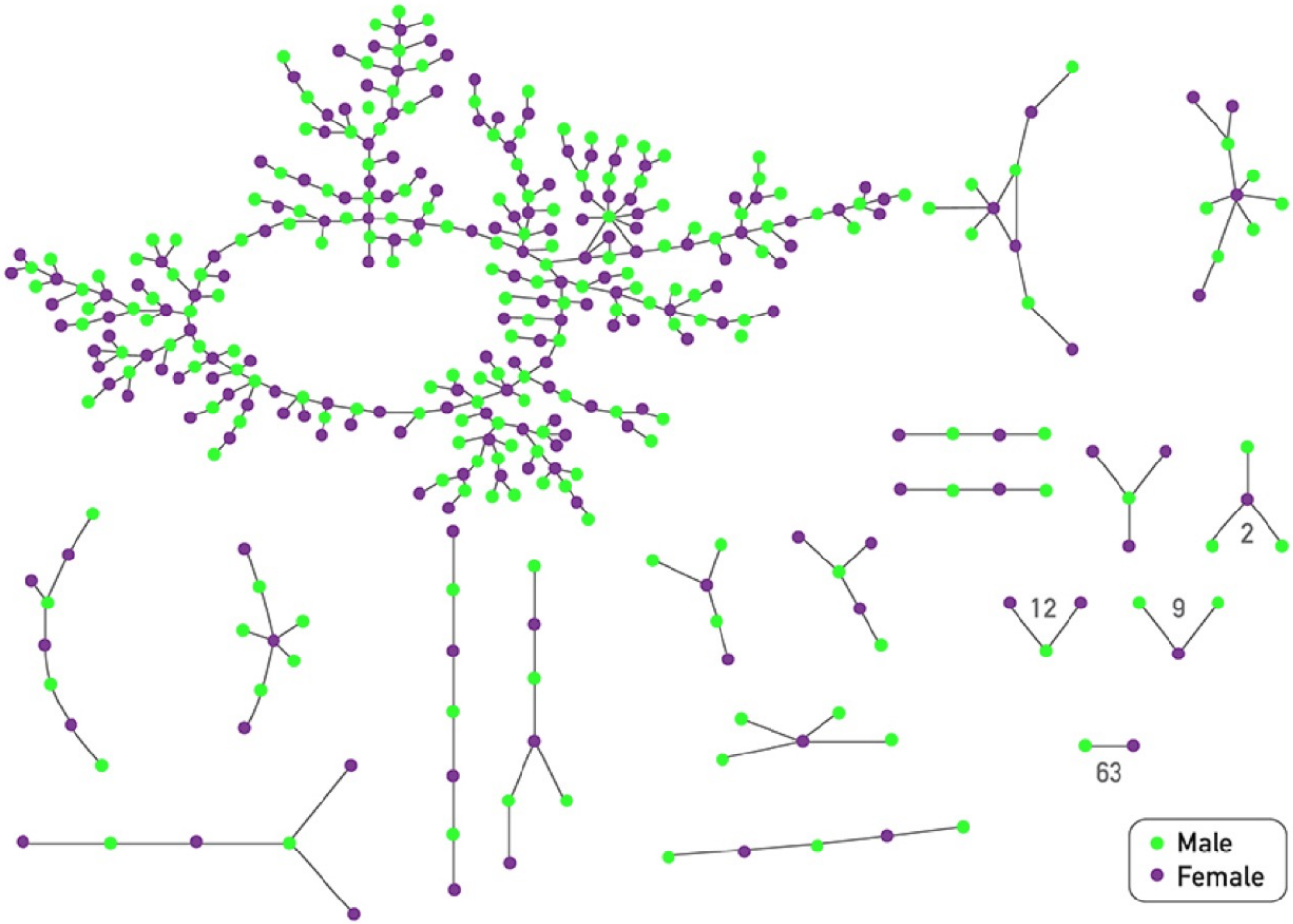
Giant Components

- The notion of giant components is also useful for networks on smaller scales
- An example is the collaboration network seen before
- Another example are the chains of affections seen before



Giant Components

- The presence of such a large component is important for the possible spreading of sexually transmitted disease
- A student may have had one single partner and -yet- being part of this large component and many paths of potential transmission



Shortest Path

- The **shortest path** between nodes *i* and *j* is the path with the fewest number of links
- The shortest path is often called **distance**, and is denoted by d_{ij}
- Undirected graph: $d_{ij} = d_{ji}$
- In a directed graph often

$$d_{ij} \neq d_{ji}$$

Diameter

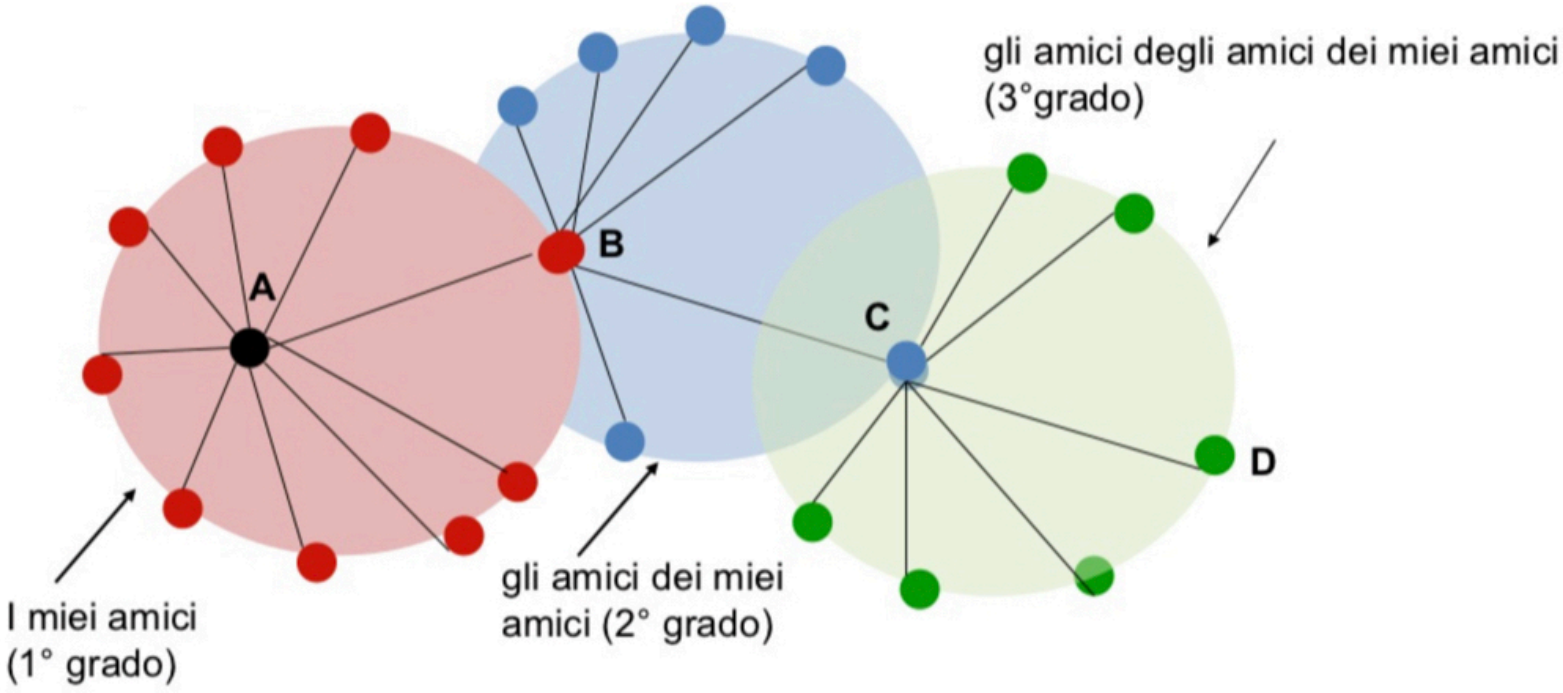
- The **diameter** of a network is the maximum shortest path in a network, d_{max}
  - That is, the largest distance recorded between any pair of nodes
- The **average path length** $\langle d \rangle$ is the average distance between all pairs of nodes in the network
- $\langle d \rangle$ is measured only for nodes that are in the same connected component

Practice on your own!

- Draw a graph
- Obtain its degree distribution
- Write its adjacency matrix
- Measure its sparsity
- Compute the distance between two nodes
- Compute the diameter of the graph
- Identify connected components

The Small-World Phenomenon

- The world looks "small" when we think of how short a path of friends it takes to get from us to almost anyone else

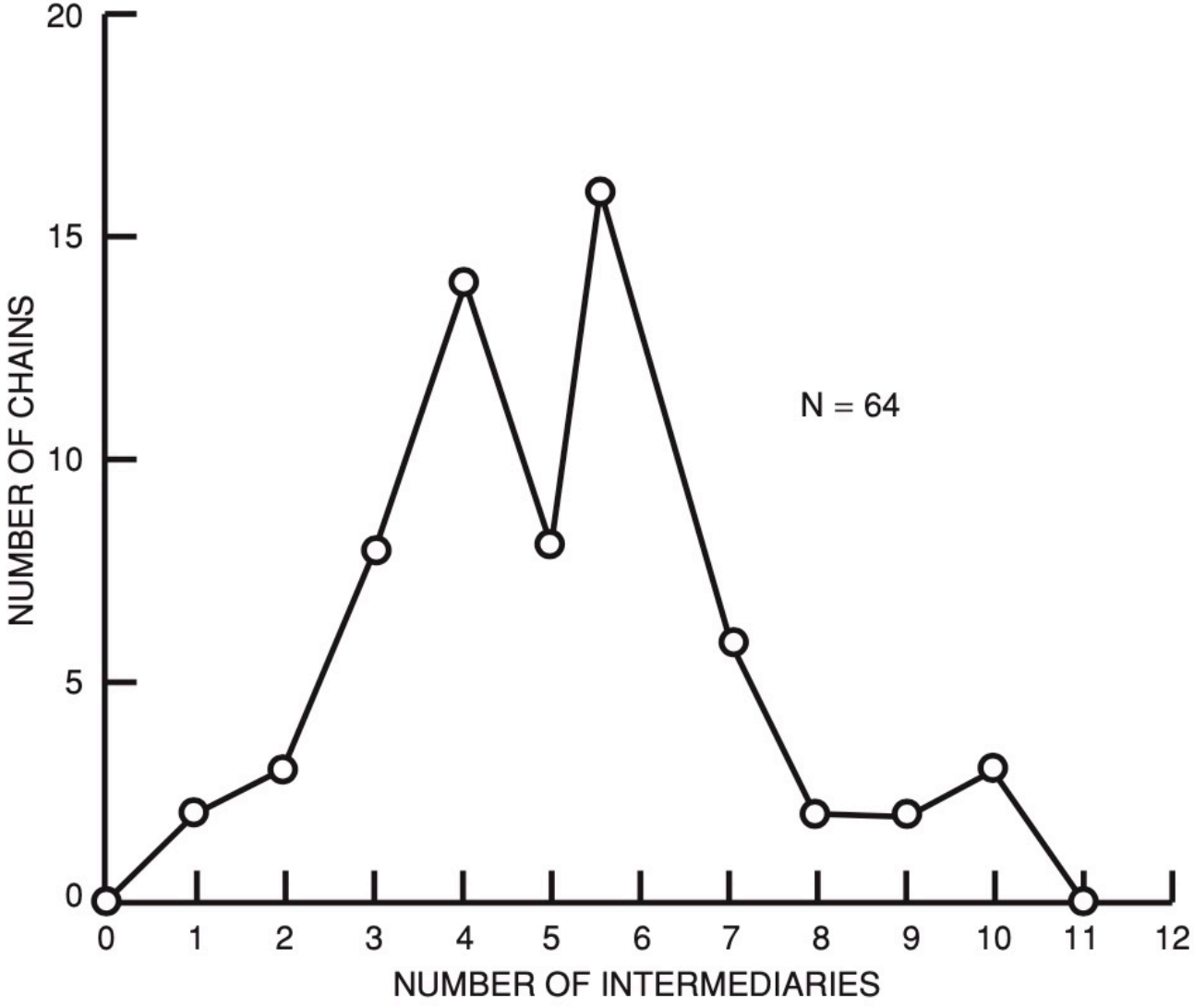


Milgram's Experiment (1967)

- How many intermediaries are necessary to connect any two people in the world?
- Milgram asked a set of randomly selected *starters* to send a letter to a certain *target* person
- Starters knew the name of the target, his working activity, and that he lived in a suburb of Boston
- Each starter was asked to forward the letter to someone they knew, with the same instructions, to eventually reach the target asap

Milgram's Experiment (1967)

- The figure below shows the distribution of path lengths among the 64 chains that reached the target



"My, it's a small world"

- The median length of chains in Milgram's experiment was six
- The Small-World Phenomenon is indeed known as the **six degrees of separation**

The simplest way of formulating the small-world problem is: Starting with any two people in the world, what is the probability that they will know each other? A somewhat more sophisticated formulation, however, takes account of the fact that while persons X and Z may not know each other directly, they may share a mutual acquaintance—that is, a person who knows both of them. One can then think of an acquaintance chain with X knowing Y and Y knowing Z. Moreover, one can imagine circumstances in which X is linked to Z not by a single link, but by a series of links, X-*a*-*b*-*c*-*d* . . . *y*-Z. That is to say, person X knows person *a* who in turn knows person *b*, who knows *c* . . . who knows *y*, who knows Z.

The Small-World Phenomenon

- Social networks tend to have very short paths between arbitrary pairs of people
- The existence of these short paths has consequences..
- Any examples?

The Small-World Phenomenon

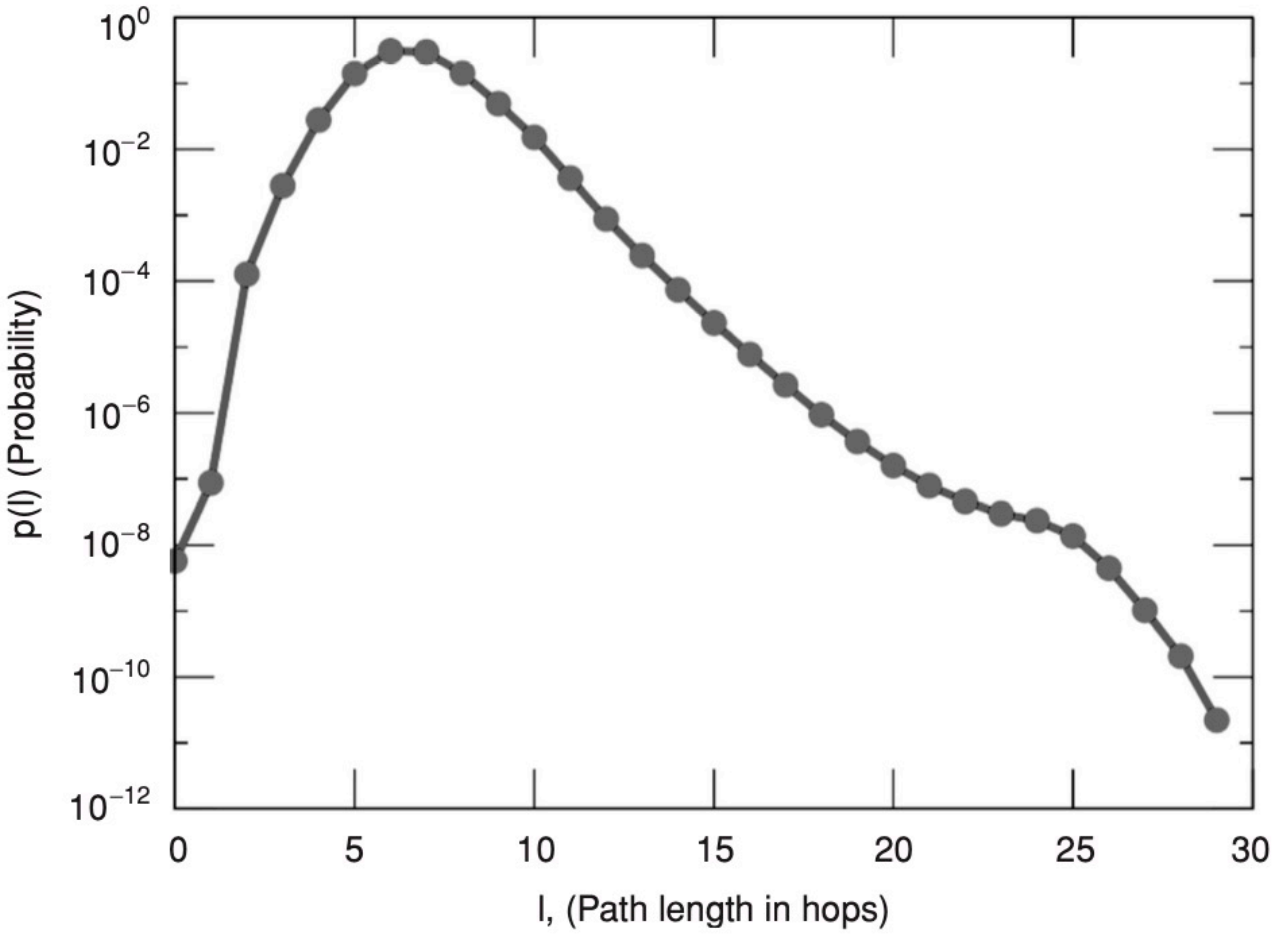
- Social networks tend to have very short paths between arbitrary pairs of people
- The existence of these short paths has consequences..
- Examples:
  - Spreading of disease, information, and any kind of social contagion

The Small-World Phenomenon

- The consensus that social networks are small worlds depends on the fact that the notion has been confirmed in settings with full data availability
- In 2008 Leskovec and Horvitz analyzed the interactions of 240M users on Microsoft Instant Messenger (IM)
- They built a graph in which a node is a user and an edge exists between two users if they have engaged in a conversation

The Small-World Phenomenon

- The graph turned out to have a giant component containing almost of all the nodes
- The distances in this component were very small: est. average 6.6, est. median 7



The Small-World Phenomenon

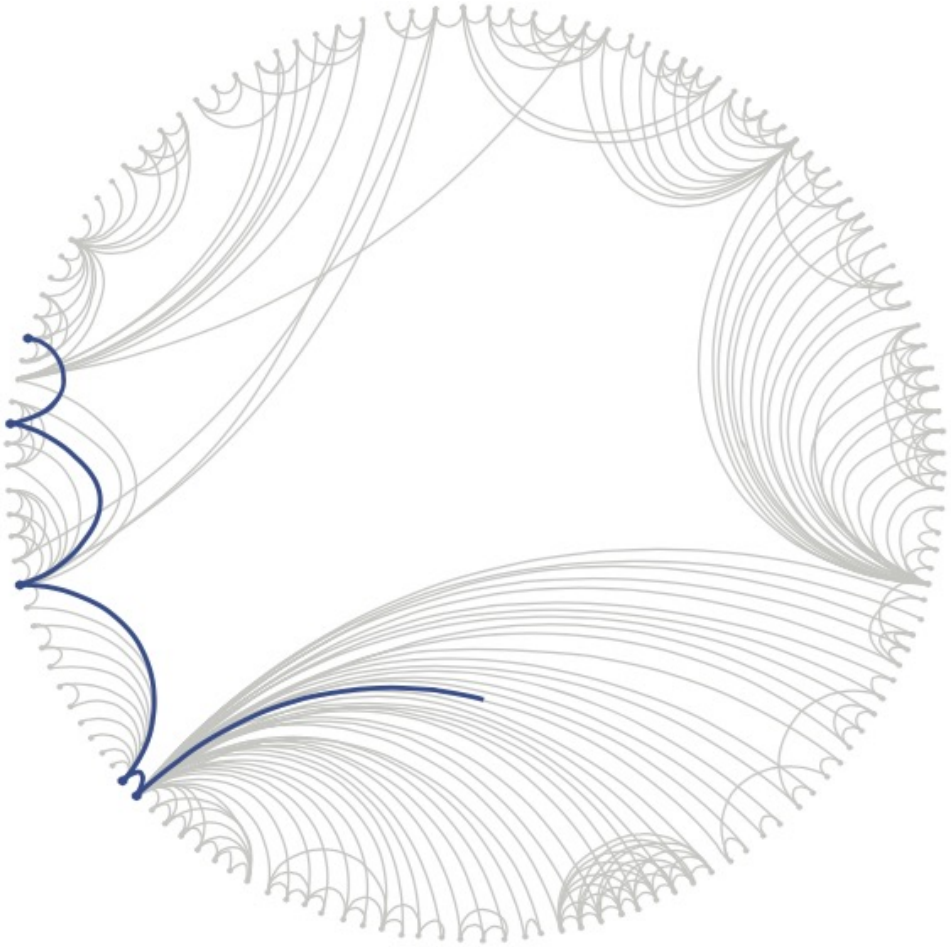
- The plot starts to approximate the distribution of how far apart we are in the global friendship network, but there are some issues..
- Any idea?

The Small-World Phenomenon

- The Microsoft study is some distance away from Milgram's goal
- The study only tracks people who access instant messaging
- The graph is not based on real friendships, but on conversations

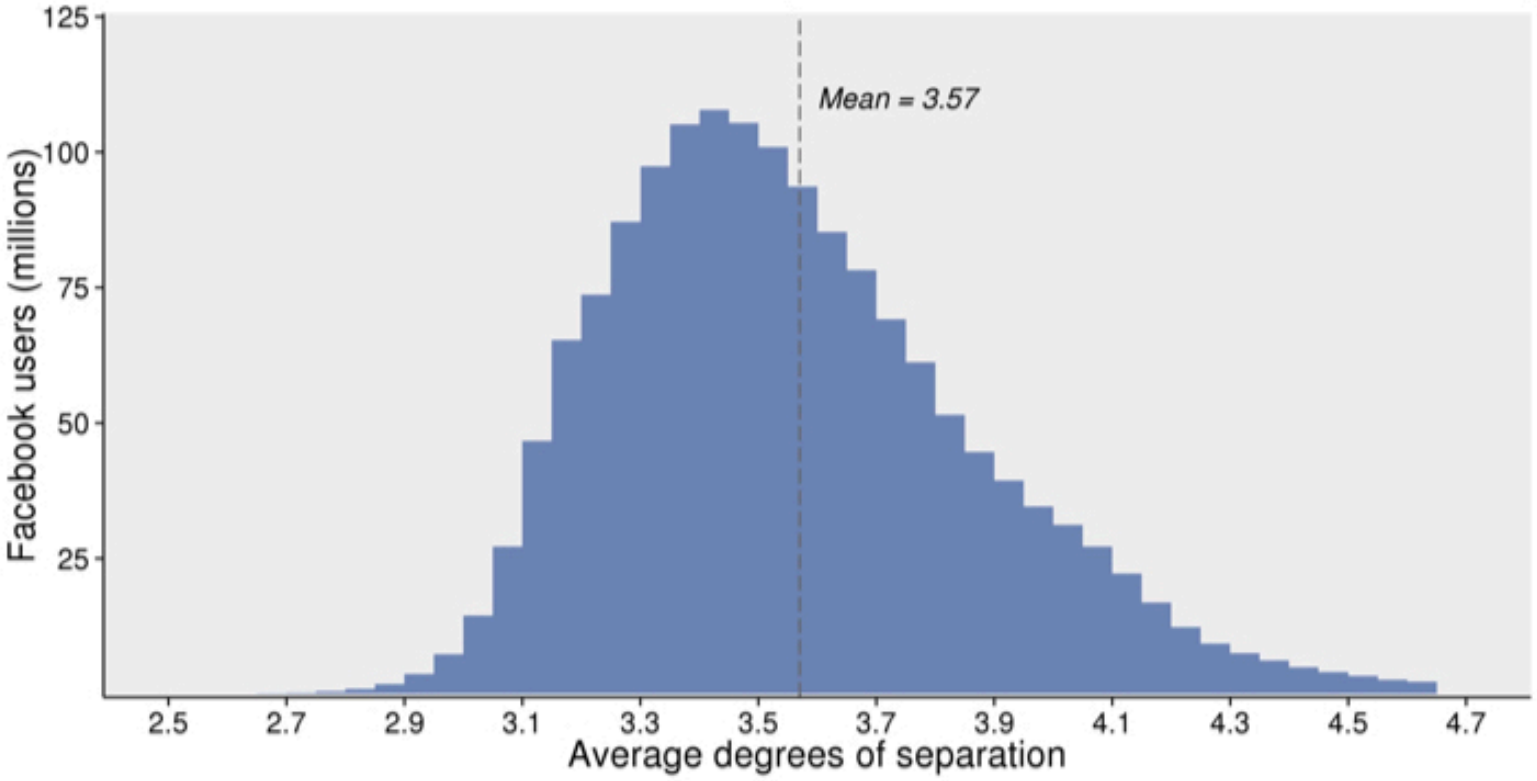
The Small-World Phenomenon

In 2016 Bhagat and colleagues estimated the average degrees of separation between all people on Facebook



The Small-World Phenomenon

- The average person is connected to every other person by an average of 3.57 steps
- The majority of people have an average between 3 and 4 steps



Notes

This presentation is based on:

- Chapter 2 of Albert Laszlo Barabasi, *Network Science*, Cambridge University Press, 2016
- Chapter 2 of David Easley and Jon Kleinberg, *Networks, crowds, and markets*, Cambridge University Press, 2012